Prompt-Based Multi-Interest Learning Method for Sequential Recommendation

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***Abstract*—Multi-interest learning method for sequential recommendation aims to predict the next item according to user multi-faceted interests given the user historical interactions. Existing methods mainly consist of a multi-interest extractor that embeds the user interactions into the user multi-interest embeddings, and a multi-interest aggregator that aggregates the learned multi-interest embeddings to the final user embedding, used for predicting the user rating to an item. Despite their effectiveness, existing methods have two key limitations: 1) they directly feed the user interactions into the multi-interest extractor and aggregator, while ignoring their different learning objectives, and 2) they merely consider the centrality of the user interactions to capture the user interests, while overlooking their dispersion. To tackle these limitations, we propose a prompt-based multi-interest learning method (PoMRec), where specific prompts are inserted into the inputted user interactions to make them adaptive to the multi-interest extractor and aggregator. Moreover, we utilize both the mean and variance embeddings of user interactions to embed the user multiple interests for the comprehensively user interest learning. We conduct extensive experiments on three public datasets, and the results verify that our proposed PoMRec outperforms the state-of-the-art multi-interest learning methods.**

***Index Terms*—Sequential recommendation, multi-interest learning method, prompt based learning.**

I. Introduction

RECOMMENDER systems have become increasingly prevalent in real-world applications, which could recommend items for users based on their interests. Their core is to learn the user and item embeddings, and then predict the user rating to the item with the distance between their embeddings. Traditional methods [1], [2], [3] treat the user historical interactions as a set, ignoring their sequential information. Intuitively, a user may purchase a phone case after buying a cellphone. Therefore, many researches [4], [5], [6], [7] formalize the sequential recommendation, where user interactions are treated as an ordered sequence to predict the next item that the user will be interested in. The mainstream sequential recommendation methods focus on devising various neural networks, such as the recurrent neural networks [4], [5] and Transformer [8], to encode the user interaction sequence into a single embedding to represent the user interests.

However, user interests are diverse and multi-faceted. For example, a girl may be simultaneously interested in jewelry, handbags, and make-ups. In such case, a single interest embedding could be hard to accurately capture all the user's diverse interests. Hence, several multi-interest learning methods [9], [10], [11], [12] for the sequential recommendation have been proposed to learn multiple interest embeddings for each user. Generally, these methods involve two key modules. 1) The *multi-interest extractor* that learns multiple embeddings for one user to capture his/her multiple interests. And 2) the *multi-interest aggregator* that aggregates the multi-interest embeddings to one user embedding based on the predicted weights. Then the user rating to one item is calculated by the dot product between the user and item embeddings. Despite their effectiveness, existing multi-interest learning methods for the sequential recommendation still have the following limitations:

- *Fail to adapt the user interactions to different learning objectives of the multi-interest extractor and aggregator:* Since both the user multi-interest embeddings and their aggregation weights can be referred to the user interactions, existing methods directly take the user interactions as the inputs to the multi-interest extractor and aggregator. In fact, the extractor focuses on the content of user interactions to embed the user multiple interests, while the aggregator emphasizes analyzing the distribution of user interactions over multiple interests to derive the aggregation weights. Therefore, existing methods fail to make the input user interactions adaptive to different learning objectives of the two modules, whereby may derive the imprecise results.
- *Overlook the dispersion of user interactions during the user multi-interest learning:* Existing approaches mainly select a representative embedding of the user interaction embeddings, e.g., the weighted summation, as one of the user multi-interest embeddings. This representative embedding could capture the centrality of user interactions. However, the user interactions can be dispersed and their dispersion also benefits the representation of the user interests [13],

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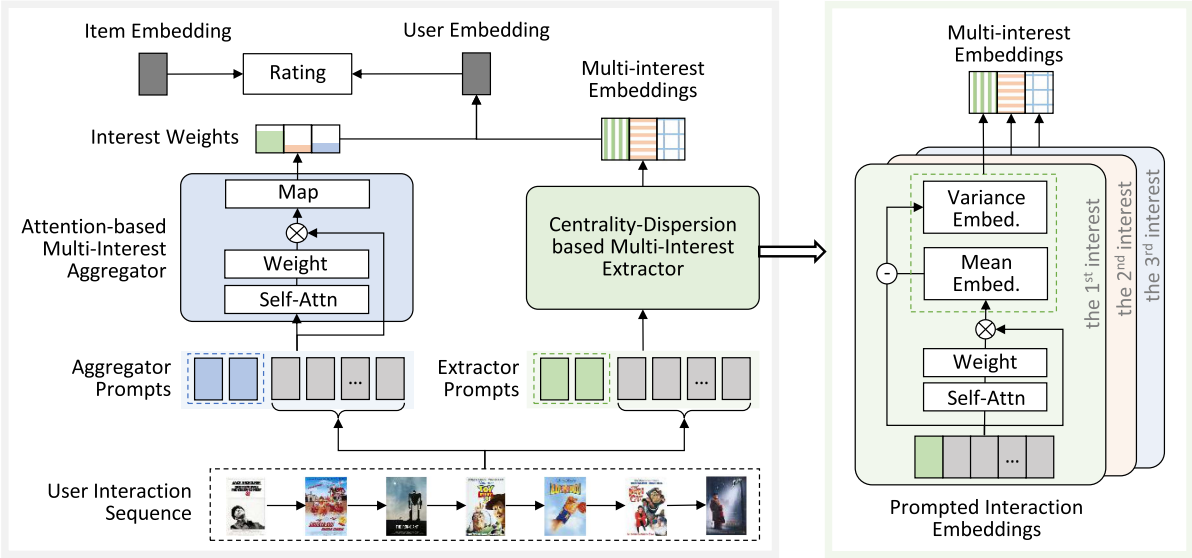

Fig. 1. Illustration of the proposed prompt-based multi-interest learning method, which consists of a centrality-dispersion based multi-interest extractor and an attention-based multi-interest aggregator. Before feeding the user interaction sequence as the input, we insert different prompts at its beginning to adapt it to the different learning objectives of the two components. Besides, in the centrality-dispersion based multi-interest extractor, we propose to capture the user multi-interests based on both the mean and variance embeddings of the user interactions.

[14]. Therefore, it is insufficient for existing methods to learn the user interests with only the centrality of user interactions.

To address the aforementioned limitations, in this paper, we propose a **Pro**mpt-based **M**ulti-interest learning method for the sequential **Rec**ommendation, termed as PoMRec, as shown in Fig. 1. Similar to existing multi-interest learning methods, PoMRec contains two modules: a multi-interest extractor and a multi-interest aggregator. Differently, inspired by the prompt engineering [15], [16] that can make the original inputs adaptive to different downstream tasks, we first introduce specific prompts for the multi-interest extractor and aggregator, respectively. As shown in Fig. 1, we insert the prompts at the front of the user interaction sequence and feed them into the downstream modules. The prompts can help each module identify that whether it should focus on the content of user interactions to derive the multi-interest embeddings or their distribution over multiple interests to derive the aggregation weights. Thereafter, the output user multi-interest embeddings are aggregated by the interest weights as the final user embedding to predict the user rating to a given item. Besides, to embed the user multiple interests in a more comprehensive manner, we propose to learn the multi-interest embeddings considering both the centrality and dispersion of the user interactions in the multi-interest extractor shown in Fig. 1. In particular, as for each user interest, we adopt the self-attention mechanism to calculate the probability weight of each user interaction belonging to this interest. Based on these weights, we calculate the mean and variance embeddings of all interactions, and incorporate them as the embedding representing this user interest. We conduct extensive experiments on three public datasets: ML-1 M, Beauty, and Movie & TV. Experimental results demonstrate the effectiveness of the proposed PoMRec.

The main contributions can be summarized as follows:

- We propose a prompt-based multi-interest learning method for the sequential recommendation (PoMRec) that introduces the learnable prompt embeddings into the input user interactions. The prompt embeddings adapt the input to the different learning objectives of the multi-interest extractor and aggregator, achieving the better recommendation performance.
- Different from existing methods that only consider the centrality of user interactions when learning the user multi-interests, we propose a centrality-dispersion based multi-interest extractor that further takes the dispersion of user interactions as a supplement for obtaining a better representation of the user multiple interests.
- Extensive experiments on the three public datasets have demonstrated the effectiveness of the proposed method. We have released our codes to facilitate other researchers in https://github.com/hello-dx/PoMRec.

II. Related Work

In this section, we first introduce the single-interest and multi-interest learning methods in Sections II-A and II-B, respectively. Besides, we review the prompt tuning methods in Section II-C.

A. Single-Interest Learning Method

The sequential recommendation aims to predict the next item that the user might be interested in based on the user historical interaction sequence [4], [6], [17], [18]. Early researches [4], [6], [19] focus on encoding the user interaction sequence into a single embedding to represent the user single-interest. For example, Cheng et al. [19] adopted Markov chains to capture the correlation among items. With the development of deep neural

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networks, several studies employ the sequential modeling techniques, e.g., the Gate Recurrent Unit [4] and Long Short-Term Memory [20], to encode the user interaction sequence into a single embedding to predict the next item. Recently, the attention mechanism [6], [21], [22] has shown promising potential to capture context-aware interests. However, user interests are diverse and multi-faceted. The aforementioned methods that utilize a single embeddings is hard to capture all the user's diverse interests. Besides, existing methods only leverage one embedding to represent the centrality of user interactions, which might be insufficient to capture the user interests when the user interactions are dispersed.

B. Multi-Interest Learning Method

As user interests are diverse, multi-interest learning has been proposed for the sequential recommendation, which aims to learn multiple embeddings for one user to capture the user multi-interests [9], [10], [23], [24], [25]. Existing methods generally involve a multi-interest extractor to learn the multiple interests of the user, and a multi-interest aggregator to fuse the multiple interests for generating the final recommendation. In particular, as for the extractor, MIND [9] uses the dynamic capsule routing method to group user interactions and obtain multiple user embeddings. Cen et al. [10] adopted the self-attention mechanism to generate multiple embeddings from the inputted user interactions. As for the aggregator, several approaches utilize the simple greedy inference strategy [10], [26] that utilizes the best matching interest embedding (among all interest embeddings) to rank the item. Other recent approaches [11], [23] predict a weight vector to aggregate the user multi-interest embeddings into the user embedding, which shows greater performance. For example, Wang et al. [11] utilized the Gate Recurrent Unit (GRU) to encode user interactions into one embedding and mapped it to the weight vector by the multi-layer perceptron.

Nevertheless, these methods directly feed the user interactions into the multi-interest extractor and aggregator, without making the inputs adaptive to the different learning objectives. In this paper, we introduce specific prompts into user interactions in order to make the downstream model known whether it should focus on the contents of user interactions or the user preference over multiple interests.

C. Prompt Tuning

The prompt learning is first proposed to overcome the gap between the pre-training and fine-tuning [16], [27]. It focuses on adding prompts to the downstream tasks of the pre-training models to improve the performance of the downstream tasks without the fine-tuning step. Early approaches [28], [29] mostly incorporate manually generated discrete prompts to guide the model. Later, since the manually generated prompts are both time-consuming and trivial, other researches [30], [31], [32] turn to automatically search discrete prompts for specific tasks. Nevertheless, these methods largely depend on the quality of the generated prompts. Some recent approaches have begun to utilize the continuous learnable embeddings as the prompts [15], [33], [34] achieving the state-of-the-art performance.

Due to the flexibility and superiority of the continuous learnable prompt embeddings over manual designed prompts, in this work, we introduce specific learnable embeddings for the multi-interest extractor and aggregator, respectively, to make the inputs adaptive to these two components. Although the prompt embeddings are learned from scratch, they are empirically optimized to capture the specific learning objectives of the multi-interest extractor and aggregator.

III. Methodology

We propose a prompt-based multi-interest learning method for sequential recommendation (PoMRec). In this section, we first brief the problem definition of the sequential recommendation in Section III-A. We then detail the proposed PoMRec in Section III-B, followed by the model complexity analysis in Section III-C.

A. Problem Definition

To improve the readability, we declare the notations used in this paper. We use the squiggled letters (*e.g.,* $\mathcal{X}$) to represent sets. The bold capital letters (*e.g.,* $\boldsymbol{X}$) and bold lowercase letters (*e.g.,* $\boldsymbol{x}$) represent matrices and vectors, respectively. Let the nonbold letters (*e.g.,* x) denote scalars.

Suppose that there is a set of users $\mathcal{U}$, and a set of items $\mathcal{I}$. Each user $u \in \mathcal{U}$ is associated with a sequence of all his/her historical interactions sorted by their corresponding interacted timestamps, denoted as $S_u = [i_{u,1}, i_{u,2}, \ldots, i_{u,N_u}]$, where $i_{u,t}$ is the interacted item at time step t and N_u is the length of the list. Different from traditional recommendation that represents the user with a given user ID, the sequential recommendation resort to a sequence of item IDs that the user historically interacted. In particular, as for a given user, the goal of the sequential recommendation takes the user interaction sequence $S_{u,t}$ truncated as the time step t as the input and predicts the next item $i_{u,t+1}$ that the user will be interested in be. The notations used in this paper are summarized in Table I.

B. Prompt-Based Multi-Interest Learning Method for the Sequential Recommendation (PoMRec)

In this subsection, we first outline the proposed PoMRec method. Then we provide an implementation of the multi-interest extractor and aggregator in PoMRec, respectively, followed by the model optimization.

1) Overall Framework: Given the user interaction sequence $S_{u,t}$, PoMRec has a multi-interest extractor that derives the user multi-interest embeddings and a multi-interest aggregator that learns the weights to fuse the multi-interest embeddings. Different from previous methods, PoMRec inserts certain prompts into the inputted user interaction sequence $S_{u,t}$ before feeding it into the multi-interest extractor and aggregator, which makes it adaptive to different learning objectives of the two modules.

Inputs: Accordingly, the inputs of PoMRec consist of two parts: the original user interactions and newly introduced prompts for the multi-interest extractor and aggregator.

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TABLE I
SUMMARY OF THE MAIN NOTATIONS

Notation	Explanation
$\mathcal{U}, \mathcal{I}$	Sets of users and items, respectively.
S_u	User u's interaction sequence.
$S_{u,t}$	User u's interaction sequence truncated at time t.
K	Predefined number of user interests.
d	Embedding size.
$\boldsymbol{e}_i \in \mathbb{R}^d$	Embedding of the item i.
$\boldsymbol{e}_{u,t} \in \mathbb{R}^d$	Embedding of the user u at time t.
$\boldsymbol{H}_{u,t} \in \mathbb{R}^{d\times M}$	Interaction embeddings $\boldsymbol{H}_{u,t} = [\boldsymbol{e}_{u,t-M}, ..., \boldsymbol{e}_{u,t}]$ of user u contains most recent M items in $S_{u,t}$.
N_p	Number of the prompt embeddings.
$\boldsymbol{P}_{\mathcal{F}} \in \mathbb{R}^{N_p\times d}$	Prompts for multi-interest extractor.
$\boldsymbol{P}_{\mathcal{G}} \in \mathbb{R}^{N_p\times d}$	Prompts for multi-interest aggregator.
$\boldsymbol{V}_{u,t} \in \mathbb{R}^{d\times K}$	Multi-interest embeddings of the user u, where the column vector $\boldsymbol{V}_u^{(k)}$ is the k-th interest embedding.
$\boldsymbol{z}_{u,t} \in \mathbb{R}^K$	Interest weights of the user u at time t.
$y_{u,t}^i$	Predicted rating of the user u to item i at time t.
λ	Trade-off parameter between the centrality and dispersion of user interactions.

- To represent the original user interactions, following most recommendation methods [3], [10], [11], we represent each item $i \in \mathcal{I}$ with a d-dimensional learnable embedding $\boldsymbol{e}_i \in \mathbb{R}^d$. Then the user historical interaction sequence $S_{u,t}$ can be transformed into an embedding sequence $\boldsymbol{H}_{u,t} = [\boldsymbol{e}_{u,t-M+1}, \ldots, \boldsymbol{e}_{u,t-1}, \boldsymbol{e}_{u,t}]$, which consists of the most recent interacted M items at the time step t. We term the embedding sequence $\boldsymbol{H}_{u,t}$ as the user interaction embeddings. Notably, if the sequence length is less than M, we repeatedly add a zero embedding to the beginning of the sequence until the length becomes M. In addition, we also add the trainable positional embeddings to each item embedding in $\boldsymbol{H}_{u,t}$ in order to make use of the sequential information.
- We then introduce the prompts for the multi-interest extractor and aggregator. Due to the flexibility and superiority of the continuous learnable prompt embeddings [15], [33], we introduce the learnable prompt embeddings $\boldsymbol{P}_{\mathcal{F}} = [\boldsymbol{p}_{\mathcal{F}}^1, \ldots, \boldsymbol{p}_{\mathcal{F}}^{N_p}]$ for the extractor, and $\boldsymbol{P}_{\mathcal{G}} = [\boldsymbol{p}_{\mathcal{G}}^1, \ldots, \boldsymbol{p}_{\mathcal{G}}^{N_p}]$ for the aggregator, both of which consist of N_p randomly initialized d-dimensional embeddings. The prompt embeddings $\boldsymbol{P}_{\mathcal{F}}$ and $\boldsymbol{P}_{\mathcal{G}}$ are learned from scratch and will be optimized to capture the specific learning objectives of the extractor and aggregator, respectively.

Formally, we insert the prompt embeddings at the beginning of the interaction embeddings $\boldsymbol{H}_{u,t}$ as the inputs for the extractor and aggregator as follows,

$$\begin{cases} \boldsymbol{H}_{u,t}^{\mathcal{F}} = [\boldsymbol{P}_{\mathcal{F}}, \boldsymbol{H}_{u,t}], \\ \boldsymbol{H}_{u,t}^{\mathcal{G}} = [\boldsymbol{P}_{\mathcal{G}}, \boldsymbol{H}_{u,t}], \end{cases} \tag{1}$$

where $\boldsymbol{H}_{u,t}^{\mathcal{F}} \in \mathbb{R}^{(N_p+M)\times d}$ and $\boldsymbol{H}_{u,t}^{\mathcal{G}} \in \mathbb{R}^{(N_p+M)\times d}$ are the prompt-augmented interaction embeddings inputted into the multi-interest extractor and aggregator, respectively.

Centrality-Dispersion based Multi-interest Extractor: Considering that the user's current interests can be inferred by his/her recent interactions, this extractor $\mathcal{F}$ will output the user multi-interest embeddings based on the prompt-augmented interaction embeddings $\boldsymbol{H}_{u,t}^{\mathcal{F}}$. Formally, suppose that each user has K interests. The user multi-interest embeddings $\boldsymbol{V}_{u,t}$ can be derived as follows,

$$\boldsymbol{V}_{u,t} = \mathcal{F}(\boldsymbol{H}_{u,t}^{\mathcal{F}} | \boldsymbol{\Theta}_{\mathcal{F}}), \tag{2}$$

where $\boldsymbol{V}_{u,t} \in \mathbb{R}^{d\times K}$ refer to the total K interest embeddings of the user u at the time step t. $\boldsymbol{\Theta}_{\mathcal{F}}$ is the set of parameters in the extractor $\mathcal{F}$. The detailed implementation of the extractor can refer to Section III-B2.

Attention-based Multi-interest Aggregator: The aggregator $\mathcal{G}$ focuses on aggregating the learned multi-interest embeddings $\boldsymbol{V}_{u,t}$ to one embedding for predicting the final user rating. In particular, the aggregation weights of different user interests can be traced from the user interactions. For example, if the user have purchased the outdoor equipment, he/she is not likely to prefer the same items in the next time. Therefore, we feed the prompt-augmented interaction embeddings $\boldsymbol{H}_{u,t}^{\mathcal{G}}$ into the aggregator $\mathcal{G}$ to learn the weights of multiple user interests. Then the final user embedding $\boldsymbol{e}_{u,t} \in \mathbb{R}^d$ at the time step t can be aggregated by the learned weights as follows,

$$\boldsymbol{e}_{u,t} = \mathcal{G}(\boldsymbol{H}_{u,t}^{\mathcal{G}}, \boldsymbol{V}_{u,t} | \boldsymbol{\Theta}_{\mathcal{G}}), \tag{3}$$

where $\boldsymbol{\Theta}_{\mathcal{G}}$ is the set of parameters in the aggregator $\mathcal{G}$. The detailed implementation can refer to Section III-B3.

Output. Ultimately, we use the commonly-used dot product between the user and item embeddings as the rating of the user u to a given item i as follows,

$$y_{u,t}^i = \boldsymbol{e}_{u,t}^{\mathsf{T}} \boldsymbol{e}_i, \tag{4}$$

where $y_{u,t}^i$ is the rating of the user u to the item i at the time step t. Accordingly, based on the predicted user ratings, the candidate items can be ranked and top items are recommended to the user.

2) Centrality-Dispersion Based Multi-Interest Extractor: The basic idea of the centrality-dispersion based multi-interest extractor is to first represent the centrality of each interest in the user interactions and then calculate the dispersion according to the centrality as the supplement. To be more specific, following studies [10], [11], we adopt the self-attention mechanism to learn the centrality representation matrix $\boldsymbol{M}_{u,t} \in \mathbb{R}^{d\times K}$ of the user multi-interests as follows,

$$\begin{cases} \boldsymbol{M}_{u,t} = (\boldsymbol{H}_{u,t}^{\mathcal{F}})^{\mathsf{T}} \boldsymbol{A}_{u,t}^{\mathcal{F}}, \\ \boldsymbol{A}_{u,t}^{\mathcal{F}} = \mathrm{softmax}\left(\boldsymbol{W}_{\mathcal{F}}^2 \tanh(\boldsymbol{W}_{\mathcal{F}}^1 \boldsymbol{H}_{u,t}^{\mathcal{F}})\right)^{\mathsf{T}}, \end{cases} \tag{5}$$

where $\boldsymbol{A}_{u,t}^{\mathcal{F}} \in \mathbb{R}^{(N_p+M)\times K}$ is an attention matrix, which represents the probability of each user interaction belonging to each interest of the user u. Intuitively, the attention matrix is able to softly classify the user interactions into K groups. Accordingly, the k-th column of the centrality representation matrix, i.e, $\boldsymbol{M}_{u,t}^{(k)}$ is derived by the weighted summation of the interaction embeddings, which represents the centrality of the k-th interests of the user u. The function $\mathrm{softmax}$ is to enforce the summation of the elements in the k-th column vector of the attention matrix

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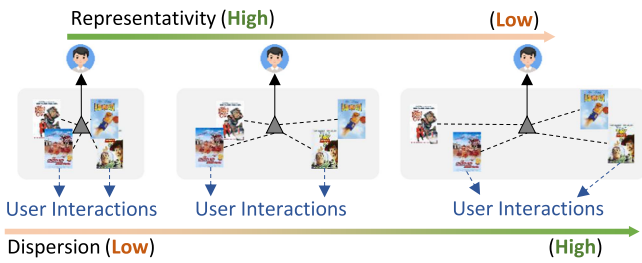

Fig. 2. Toy illustration of the user interactions with different dispersion. The grey triangles represent the centrality of the user interactions. Intuitively, as the dispersion of user interactions increasing, the representativity of the centrality in capturing the user interactions decreases.

to equal to 1. $\boldsymbol{W}_{\mathcal{F}}^1 \in \mathbb{R}^{d' \times d}$ and $\boldsymbol{W}_{\mathcal{F}}^2 \in \mathbb{R}^{K \times d'}$ are the trainable parameters. $\tanh$ is the activation function. It is worth noting that other approaches of learning the centrality representation matrix $\boldsymbol{M}_{u,t}$are avaliable.

Existing approaches directly utilize the centrality representation matrix $\boldsymbol{M}_{u,t}$ to represent the user interests. Nevertheless, as shown in Fig. 2, along with the user interactions becoming increasingly dispersed, the representativity of the centrality (see the grey triangles) for the user interactions tends to decrease. To tackle this issue, in this work, we first calculate a dispersion representation matrix based on the centrality representation matrix, and then derive the use multi-interest embeddings by their combination. Technically, we define the multi-interest embeddings $\boldsymbol{V}_{u,t} \in \mathbb{R}^{d \times K}$ of the user u as follows,

$$\begin{cases} \boldsymbol{V}_{u,t} = \boldsymbol{M}_{u,t} + \lambda \boldsymbol{\Sigma}_{u,t}, \\ \boldsymbol{\Sigma}_{u,t}^{(k)} = \operatorname{sqrt}\left(((\boldsymbol{H}_{u,t}^{\mathcal{F}})^2)^\top \boldsymbol{A}_{u,t}^{\mathcal{F}(k)} - (\boldsymbol{M}_{u,t}^{(k)})^2 \right), k = 1, \ldots, K, \end{cases} \tag{6}$$

where $\boldsymbol{\Sigma}_{u,t} \in \mathbb{R}^{d \times K}$ is the dispersion representation matrix, calculated by the the dispersion of user interactions from each centrality. λ is a hyper-parameter to adjust the trade-off between the centrality and dispersion of the user interactions in learning the user multi-interests. The superscript ^{2} of a matrix denotes the element-wise multiplication of the matrix. $\boldsymbol{M}_{u,t}^{(k)}$ is the k-th column vector of the matrix $\boldsymbol{M}_{u,t}$.

3) Attention-Based Multi-Interest Aggregator: The aggregator $\mathcal{G}$ first predicts a weight vector that captures the weights of user multiple interests, and then aggregates the user multi-interest embeddings $\boldsymbol{V}_{u,t}$ to the final user embedding. To fulfill this, we first aggregate these embeddings into a single embedding, and then map the embedding into a K-dimensional vector as the weight vector.

Technically, due to the impressive ability of self-attention mechanism in learning the attention weight of the embedding aggregation, we adopt it to aggregate the prompt-augmented interaction embeddings $\boldsymbol{H}_{u,t}^{\mathcal{F}}$ to derive a summarized embedding $\boldsymbol{v}_{u,t} \in \mathbb{R}^d$ as follows,

$$\begin{cases} \boldsymbol{a}_{u,t}^{\mathcal{G}} = \operatorname{softmax}\left(\boldsymbol{W}_{\mathcal{G}}^2 \tanh(\boldsymbol{W}_{\mathcal{G}}^1 \boldsymbol{H}_{u,t}^{\mathcal{G}}) \right)^\top, \\ \boldsymbol{v}_{u,t} = \boldsymbol{H}_{u,t}^{\mathcal{G}} \boldsymbol{a}_{u,t}^{\mathcal{G}}, \end{cases} \tag{7}$$

where $\boldsymbol{a}_{u,t}^{\mathcal{G}} \in \mathbb{R}^{N_p + M}$ contains the attention weights of the prompt-augmented interaction embeddings to derive the summarized embedding. $\boldsymbol{W}_{\mathcal{G}}^1 \in \mathbb{R}^{d' \times d}$ and $\boldsymbol{W}_{\mathcal{G}}^2 \in \mathbb{R}^{1 \times d'}$ are the trainable parameters.

We then adopt the multi-layer perceptron to map the summarized embedding to the weight vector as follows,

$$\boldsymbol{z}_{u,t} = \underbrace{\cdots \boldsymbol{W}_M^2 \tanh(\boldsymbol{W}_M^1}_{L} \boldsymbol{v}_{u,t} + \boldsymbol{b}_M^1) + \boldsymbol{b}_M^2 \cdots, \tag{8}$$

where $\boldsymbol{z}_{u,t} \in \mathbb{R}^K$ denotes the weights of multiple interests of the user u at the time step t. L is the layer of the multi-layer perceptron, which is set to 2 empirically. $\boldsymbol{W}_M^1 \in \mathbb{R}^{d' \times d}$, $\boldsymbol{b}_M^1 \in \mathbb{R}^{d'}$, $\boldsymbol{W}_M^2 \in \mathbb{R}^{d' \times K}$ and $\boldsymbol{b}_M^2 \in \mathbb{R}^K$ are the trainable parameters.

Accordingly, we aggregate the learned user multi-interest embeddings $\boldsymbol{V}_{u,t}$ based on the weight vector $\boldsymbol{z}_{u,t}$ to derive the user embedding as follows,

$$\boldsymbol{e}_{u,t} = \boldsymbol{V}_{u,t} \boldsymbol{z}_{u,t}, \tag{9}$$

where $\boldsymbol{e}_{u,t} \in \mathbb{R}^d$ captures the user multi-interests at the time step t, which is used to predict the final user rating for the recommendation.

4) Model Optimization: The goal of the sequential recommendation is to predict the next item given the user interaction sequence at one time step. Accordingly, following Bayesian personalized ranking mechanism [35], we build the following training set,

$$\mathcal{D} = \left\{ (S_{u,t}, i, j) \;\middle|\; \begin{array}{l} u \in \mathcal{U}, t = 1, \ldots, N_u - 1, \\ i = i_{u,t+1} \in S_u, j \in \mathcal{I} \setminus S_u \end{array} \right\} \tag{10}$$

where $S_{u,t}$ is the user interaction sequence truncated at time step t, $i = i_{u,t+1}$ is the target next item in S_u, and $j \in \mathcal{I} \setminus S_u$ is a negative item that randomly sampled from items that the user has not interacted. The training triplet $(S_{u,t}, i, j)$ indicates that the user u prefers item i compared to the item j at the time step t. We then adopt the pair-wise objective function to ensure the rating of the user u to the positive item i is larger than the negative item j as follows,

$$\mathcal{L} = \min_{\boldsymbol{\Theta}} - \sum_{\mathcal{D}} \log \left(\operatorname{sigmoid}(y_{u,t}^i - y_{u,t}^j) \right), \tag{11}$$

where $y_{u,t}^i$ and $y_{u,t}^j$ are the ratings of the user u to the item i and j at the time step t, which can be derived in (4). $\operatorname{sigmoid}$ is the sigmoid activation function. $\boldsymbol{\Theta}$ is the set of all parameters in the method, consisting of the item embeddings $\{\boldsymbol{e}_i\}_{i \in \mathcal{I}}$, the prompt embeddings $\boldsymbol{P}_{\mathcal{F}}$ and $\boldsymbol{P}_{\mathcal{G}}$, and parameters in the multi-interest extractor $\boldsymbol{\Theta}_{\mathcal{F}}$ and target-interest predictor $\boldsymbol{\Theta}_{\mathcal{G}}$. The detailed training process of the proposed PoMRec is summarized in Algorithm 1.

C. Model Complexity Analysis

In PoMRec, the main time-consuming steps appear in the user multi-interest extraction and multi-interest weight prediction, i.e., step 6 and step 7 in Algorithm 1.

- Step 6 in Algorithm 1 learns the user multi-interest embeddings, and it has computational complexity $\mathcal{O}((dd' + d'K)(M + N_p))$ to derive the attention matrix $\boldsymbol{A}_{\mathcal{F}u}$. Based on the attention matrix, we then calculate the mean and variance representations of the user multi-interests, both

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Algorithm 1: Training Process of the Proposed PoMRec.

Input: The set of users $\mathcal{U}$ and set of items $\mathcal{I}$. The historical interactions S_u of each $u \in \mathcal{U}$. The training set $\mathcal{D}$. The hyper-parameters K, N_P, and λ.
Output: The model parameters $\boldsymbol{\Theta}$.
1: Randomly initialize the embeddings $\{\boldsymbol{e}_i\}_{i\in\mathcal{I}}, \boldsymbol{P}_{\mathcal{F}}, \boldsymbol{P}_{\mathcal{G}}$ and parameters $\boldsymbol{\Theta}_{\mathcal{F}}, \boldsymbol{\Theta}_{\mathcal{G}}$.
2: Shuffle the training triplets $(S_{u,t}, i, j)$ in $\mathcal{D}$.
3: **while** not converged **do**
4: Draw a mini-batch of training triplet $(S_{u,t}, i, j)$.
5: Construct the prompt-augmented user interaction embeddings $\boldsymbol{H}^{\mathcal{F}}_{u,t}$ and $\boldsymbol{H}^{\mathcal{G}}_{u,t}$.
6: Calculate the user multi-interest embeddings: $\boldsymbol{V}_{u,t} = \mathcal{F}(\boldsymbol{H}^{\mathcal{F}}_{u,t}|\boldsymbol{\Theta}_{\mathcal{F}})$.
7: Calculate the final user embedding: $\boldsymbol{e}_{u,t} = \mathcal{G}(\boldsymbol{H}^{\mathcal{G}}_{u,t}, \boldsymbol{V}_{u,t}|\boldsymbol{\Theta}_{\mathcal{G}})$.
8: Calculate the user ratings to the item i and j: $y^i_{u,t} = \boldsymbol{e}^{\top}_{u,t}\boldsymbol{e}_i, y^j_{u,t} = \boldsymbol{e}^{\top}_{u,t}\boldsymbol{e}_j$.
9: Update the parameters of PoMRec: $\boldsymbol{\Theta} \leftarrow \boldsymbol{\Theta} - \eta\frac{\partial\mathcal{L}}{\partial\boldsymbol{\Theta}}$
10: **end while**
11: **return** Model parameters $\boldsymbol{\Theta}$.

of which have the computational complexity $\mathcal{O}(dK(M + N_p))$. Accordingly, the user multi-interest embeddings can be calculated through (6) with the computational complexity $\mathcal{O}(d(M + N_p))$.

- Step 7 in Algorithm 1 learns the weight vector and it first derives the attention matrix $\boldsymbol{A}_{\mathcal{G}u}$, which has the computational complexity $\mathcal{O}(dd'(M + N_p))$. The weight vector then can be derived by the multi-layer perceptron taking the computational complexity $\mathcal{O}(dd' + d'K)$.

Therefore, the overall complexity for evaluating PoMRec is $\mathcal{O}(d'dK(M + N_p))$. Notably, compared with existing multi-interest learning methods, the proposed PoMRec only additionally introduces N_p prompt embeddings for multi-interest extraction and multi-interest weight prediction, respectively, which burdens little to the space complexity.

IV. Experiments

In this section, we first detail the experimental settings in Section IV-A, and then present the experiment results to answer the following research questions:

RQ1: How does the proposed PoMRec method perform?
RQ2: How effective are the different components?
RQ3: How do hyper-parameters affect model performance?
RQ4: Does the dispersion of user interactions help better learn the user multi-interests?
RQ5: Do the prompt embeddings successfully capture the different learning objectives?

A. Experimental Settings

Datasets: To evaluate our proposed PoMRec in the task of the top-N item recommendation, we conducted extensive experiments on the following three public datasets: *ML-1M* [36], *Bueaty* [37] and *Movies & TV* [37], which have various scales and sparsities. All these datasets contain the users' ratings on items, and each rating is annotated with its time step. For the fair comparison, we closely followed the data pre-processing of the study [11]. The statistics of the datasets are summarized in Table II.

TABLE II
Statistics of Datasets

Dataset	#User $\lvert\mathcal{U}\rvert$	#Item $\lvert\mathcal{I}\rvert$	#Interaction $\sum_u N_u$	Density $\frac{\sum_u N_u}{\lvert\mathcal{U}\rvert\times\lvert\mathcal{I}\rvert}$
ML-1M	6,040	3,706	1,000,209	4.47%
Beauty	22,363	12,101	198,502	0.07%
Movies & TV	123,960	50,052	8,765,568	0.14%

Evaluation Protocols: We adopt the leave-one-out evaluation to evaluate model performance, which is widely-used in previous studies [6], [11]. In particular, for each user in the dataset, the most recent interaction is used for testing, the second most recent interaction is used for validation, and the remaining interactions are for training. In the validation and testing, the candidate items consist of one ground-truth item and 999 randomly sampled items that the user has not interacted with. This sampled metric is shown to be comparable to using the whole set of items as candidate items [11], [38], while largely saving the computational time. We adopted Recall@N and NDCG@N to evaluate the effectiveness of our proposed PoMRec in the top-N recommendation. By default, we set $N = 5, 10, 20$.

Implementation Configuration: We set the embedding size d to 64 and length M to 20 following existing studies [10], [11], [26]. We tuned the hyper-parameters, i.e., the number of user interests K, the number of prompt embeddings N_p and the trade-off parameter λ from $\{1, 2, 3, 4, 5\}$, respectively. The results will be discussed in Section IV-D. We optimized the proposed PoMRec with Adam optimizer [39]. We set the mini-batch sizes to 256 for all the datasets. We trained the network with 200 epochs with the early stop strategy, and selected the best model according to the performance of the validation set. Typically, 200 epochs are sufficient for PoMRec to converge.

B. Performance Comparison (RQ1)

To quantitatively examine the effectiveness of the proposed PoMRec, we compared it with the following multi-interest learning methods for the sequential recommendation:

- *GRU4Rec* [4] is the first sequential recommendation method that utilizes GRU to model the user interactions into one user interest embedding.
- *MIND* [9] uses the dynamic routing mechanism in capsule networks to group the user interacted items into multiple clusters to learn the user multi-interests. It utilizes the maximal matching score between the items and the user interests as the final rating of the user to the item, which is referred to the greedy inference strategy.
- *ComiRec* [10] uses multi-head attention mechanisms to learn multiple embeddings for each user to capture their

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TABLE III
EXPERIMENTAL RESULTS OF THE PERFORMANCE COMPARISON

Dataset	Method	Recall@5	Recall@10	Recall@20	NDCG@5	NDCG@10	NDCG@20	Param.
ML-1M	GRU4Rec*	0.2730	0.3964	0.5323	0.1875	0.2273	0.2616	0.3M
	MIND*	0.1863	0.2881	0.4152	0.1229	0.1558	0.1877	0.2M
	ComiRec*	0.2513	0.3659	0.4937	0.1708	0.2078	0.2400	0.2M
	MINER	0.2758	0.3901	0.3901	0.1878	0.2246	0.2574	0.2M
	TiMiRec*	0.3091	0.4310	0.5625	0.2136	0.2529	0.2861	0.5M
	PoMRec	**0.3151**	**0.4422**	**0.5752**	**0.2188**	**0.2598**	**0.2933**	0.2M
Relative Improvement		1.94%	2.60%	2.26%	2.43%	2.73%	2.52%	-
Beauty	GRU4Rec*	0.1072	0.1552	0.2107	0.0719	0.0873	0.1013	0.9M
	MIND*	0.1193	0.1727	0.2492	0.0809	0.0981	0.1173	0.8M
	ComiRec*	0.1257	0.1832	0.2543	0.0852	0.1038	0.1217	0.8M
	MINER	0.1224	0.1740	0.2408	0.0841	0.1008	0.1176	0.8M
	TiMiRec*	0.1437	0.2006	0.2645	0.1006	0.1118	0.1350	1.6M
	PoMRec	**0.1448**	**0.2028**	**0.2683**	**0.1007**	**0.1195**	**0.1360**	0.8M
Relative Improvement		0.77%	1.10%	1.44%	-	6.88%	0.74%	-
Movies & TV	GRU4Rec	0.1928	0.2811	0.3871	0.1299	0.1584	0.1851	3.3M
	MIND	0.2394	0.3290	0.4282	0.1655	0.1954	0.2205	3.2M
	ComiRec	0.2411	0.3291	0.4275	0.1687	0.1970	0.2219	3.2M
	MINER	0.2467	0.3398	0.4426	0.1702	0.2002	0.2262	3.2M
	TiMiRec	0.2506	0.3412	0.4386	0.1755	0.2048	0.2294	6.4M
	PoMRec	**0.2747**	**0.3684**	**0.4714**	**0.1944**	**0.2246**	**0.2507**	3.2M
Relative Improvement		9.62%	7.97%	7.48%	10.77%	9.29%	8.67%	-

The best and second best results are in bold and underlined, respectively. The parameter numbers of the methods are listed in the "Param." column. The superscript * denotes that the results of the method are referred to the study [11].

diverse interests. It also utilizes the greedy inference strategy to generate the final recommendation.

- *MINER* [12] introduces a novel poly attention scheme to learn the user multi-interest embeddings. It also leverages a disagreement regularization to improve the poly attention, which enlarges the distance among different interest embeddings during training. The learned multi-interest embeddings are fed into the mean aggregation to derive the user embedding.
- *TiMiRec* [11] applies the self-attention mechanism to the multi-interest extractor to learn the user's multi-interests and adopts the GRU to learn the weight vector from the user interactions. This method is a two-stage method, which first pre-trains the multi-interest extractor with the greedy inference strategy and then fine-tunes the interest weight predictor to learn the user embedding for predicting the recommended items.

Table III shows the comparison results of all baselines and the proposed PoMRec, where the best results are in bold and the second-best results are underlined. Besides, we also listed the total parameters of each method in Table III. From Table III, we have the following observations:

1) Our proposed PoMRec achieves the best performance with respect to all metrics on all datasets. This demonstrates the effectiveness of our proposed method. The reasons behind this may be as follows: (i) by adding the specific prompts of learning objectives, the inputs of the user interactions are adaptive to the multi-interest extractor and aggregator, thus that the multi-interest embeddings and their weights can be better learned; and (ii) we utilize both centrality and dispersion of the user interactions to learn the user multi-interest embeddings, which comprehensively models the user multi-interests.
2) The proposed PoMRec not only achieves the better performance but also needs significantly fewer parameters compared to the best baseline, i.e., TiMiRec. This is because that TiMiRec allocates two embeddings for each item, with one embedding used for the multi-interest extractor and another used for the multi-interest aggregator. In contrast, the proposed PoMRec only allocates one embedding for each item, and inserts specific prompt embeddings into the item embeddings in the user interaction sequence to make them adaptive to both the extractor and aggregator.
3) Multi-interest learning methods MINER, TiMiRec, and TiMiRec perform better than the multi-interest learning methods MIND and ComiRec. This may be due to that although MIND and ComiRec learn the user multi-interests, they merely generate recommendations based on the best matching interest of the target item. This suggests that it is helpful to predict the next items that the user might be interested in by leveraging all the learned user interests. Moreover, MINER performs worse than TiMiRec and our proposed PoMRec. This may be because that MINER only utilize the mean aggregation of the learned user multi-interest embeddings to retrieve items, while TiMiRec and PoMRec utilize networks to predict interest weights to aggregate the learned user multi-interest embeddings.
4) Note that introducing the prompt embeddings could involve additional parameters, but can be negligible compared to the whole model parameters. Specifically, the model parameters largely depends on the number of items.

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TABLE IV
EXPERIMENTAL RESULTS OF THE ABLATION STUDY ON POMREC

Dataset	Variant	Recall@5	Recall@10	NDCG@5	NDCG@10
ML-1M	Base	0.3068	0.4248	0.2101	0.2484
	+Prompt	0.3129	0.4333	0.2148	0.2536
	+Disp.	0.3108	0.4320	0.2151	0.2544
	PoMRec	**0.3151**	**0.4422**	**0.2188**	**0.2598**
Beauty	Base	0.1193	0.1726	0.0823	0.0996
	+Prompt	0.1299	0.1838	0.0888	0.1061
	+Disp.	0.1293	0.1811	0.0889	0.1057
	PoMRec	**0.1448**	**0.2028**	**0.1007**	**0.1195**
Movies & TV	Base	0.2429	0.3295	0.1704	0.1983
	+Prompt	0.2498	0.3406	0.1738	0.2031
	+Dis.	0.2591	0.3536	0.1801	0.2106
	PoMRec	**0.2747**	**0.3684**	**0.1944**	**0.2246**

There are 3,706, 12,101, and 50,052 items in ML-1 M, Beauty and Movies & TV datasets, respectively. However, there are at most 10 prompt embeddings introduced, i.e., there are two tasks in our method and we assign at most 5 embeddings for each task. Therefore, the new involved parameters only account for of 0.26%, 0.06%, and 0.02% of original model parameters in ML-1 M, Beauty and Movies & TV datasets, respectively.

C. Ablation Study (RQ2)

In this subsection, to comprehensively evaluate the components in PoMRec, we first conducted ablation study on the variants of PoMRec and then tested the potential for the deployment on existing methods.

1) Comparison of Variants: To evaluate the components in our proposed PoMRec: the prompt-based learning framework and the centrality-dispersion based mutli-interest extractor, we designed the following variant methods:

- *Base:* We removed both the components in the proposed PoMRec. Specifically, we directly fed the user interaction embeddings $\boldsymbol{H}_{u,t}$ without the prompts embeddings to both the multi-interest extractor and aggregator. Besides, in the multi-interest extractor, we only utilized the centrality representation as the user multi-interest embeddings, i.e., $\boldsymbol{V}_{u,t} = \boldsymbol{M}_{u,t}$, without the dispersion representation $\boldsymbol{\Sigma}_{u,t}$.
- *+Prompt:* This variant adds prompt-based multi-interest learning method into the Base variant. Specifically, we fed the prompt-augmented user interaction embeddings $\boldsymbol{H}^{\mathcal{F}}_{u,t}$ and $\boldsymbol{H}^{\mathcal{G}}_{u,t}$ into the multi-interest extractor and aggregator, respectively.
- *+Disp:* Compared to the Base variant, this variant adopts the proposed centrality-dispersion based multi-interest extractor to derive the user multi-interest embeddings, i.e., $\boldsymbol{V}_{u,t} = \boldsymbol{M}_{u,t} + \lambda \boldsymbol{\Sigma}_{u,t}$.

The experimental results of the ablation study are displayed in Table IV. Without losing generality, we reported the Recall@5, Recall@10, NDCG@5, and NDCG@10 of the methods. As can be seen, first, the +Prompt variant outperforms the Base variant, when incorporating the prompt embeddings. This demonstrates that it is helpful to add specific prompts for the user interaction embeddings to guide their learning objectives in the multi-interest extractor and aggregator. Second, the +Disp. variant outperforms the Base variant, which proves that it is beneficial to learn the user multi-interest embedding from both the centrality and dispersion of the user interactions. Third, the two variants perform similarly in ML-1 M and Beauty datasets, while +Disp. outperforms +Prompt in Movies & TV. This may be because that the user interests in Movies & TV dataset are more dispersed than those in ML-1 M and Beauty datasets, whereby the variant that involves the dispersion of user interactions performs better.

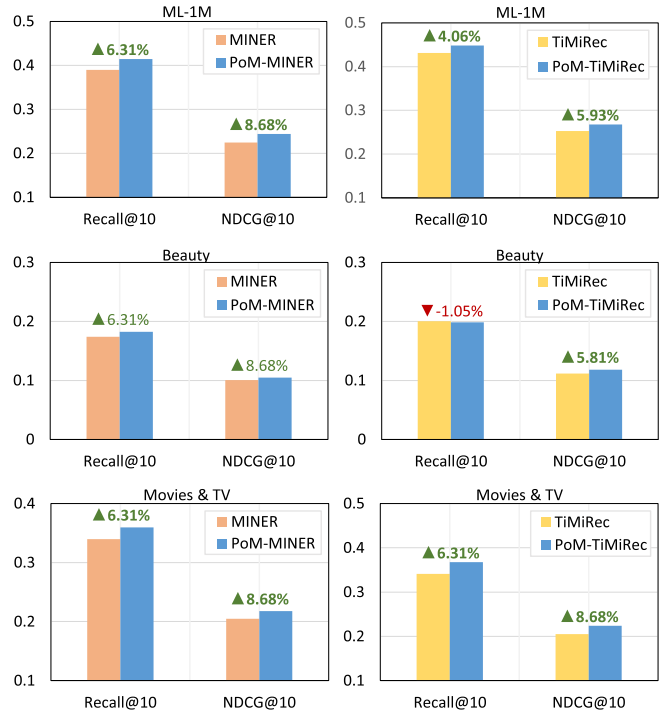

Fig. 3. Deployments of the PoMRec in existing multi-interest learning methods. We annotate the relative improvements of our deployment methods compared to the original methods.

2) Deployment on Other Methods: To test the potential of our proposed method, we deployed our proposed two components to two state-of-the-art multi-interest learning method, i.e., MINER [12] and TiMiRec [11]. In particular, we directly adopted the model architectures of the multi-interest extractor and aggregator in MINER and TiMiRec. Differently, we added the learnable prompt embeddings into the inputs of the two modules and learned the user multi-interest embeddings with both the mean and variance embeddings of user interests, as same as our proposed PoMRec. The deployments of PoMRec on MINER and TiMiRec are termed as PoM-MINER and PoM-TiMiRec, respectively. Without losing generality, we reported the Recall@10 and NDCG@10 in the three datasets in Fig. 3.

As can be seen, the deployments PoM-MINER and PoM-TiMiRec outperform their corresponding methods MINER and TiMiRec with respect to most metrics, which demonstrates the effectiveness and applicability of the proposed method. Notably, as aforementioned, the original TiMiRec

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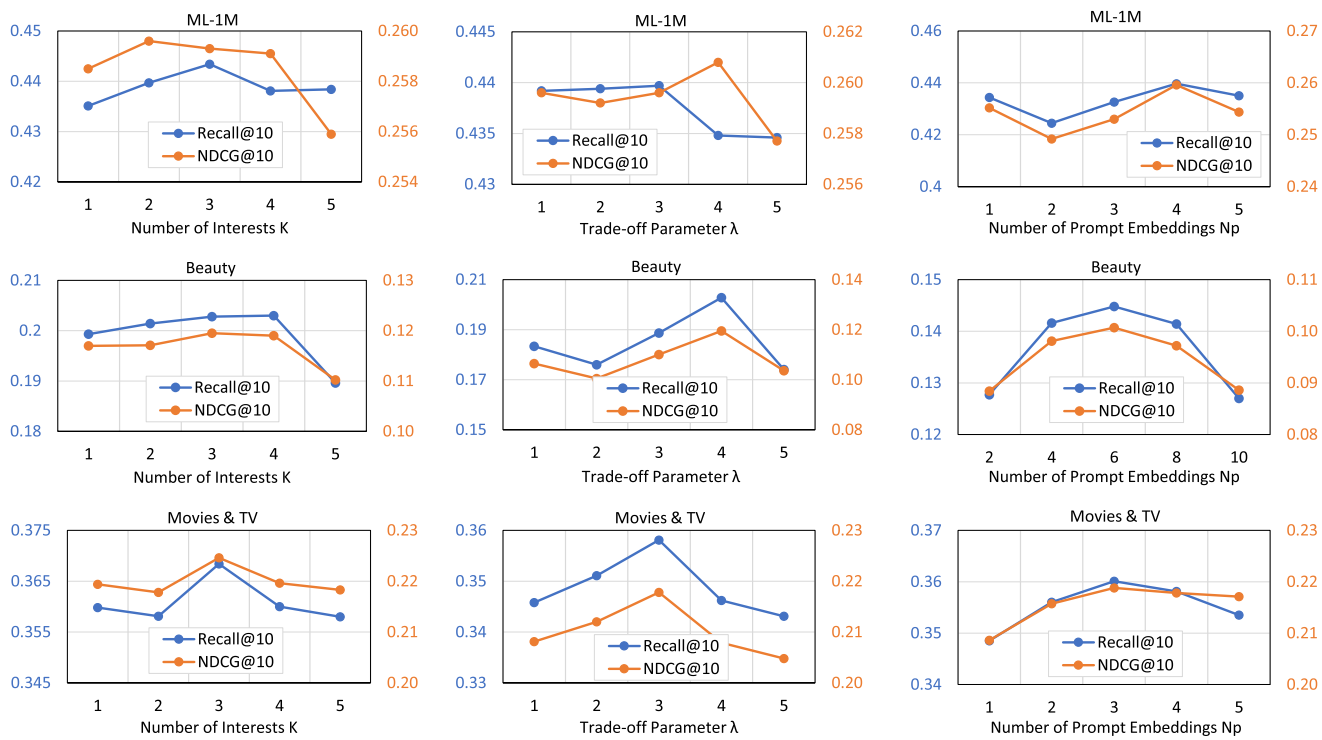

Fig. 4. Performance of different hyper-parameters. The left vertical axis refers to the Recall@10, while the right vertical axis refers to the NDCG@10.

assigns two embeddings for each item, i.e., one for the multi-interest extractor and the other one for the multi-interest aggregator. Nevertheless, our method aims to adapt the same inputs to different modules. Therefore, to deploy our method in PoM-TiMiRec, we only assigned a single embedding for each item. Intuitively, this modification could hurt the performance of PoM-TiMiRec as it obviously reduces the number of parameters. However, equipped the our proposed method, PoM-TiMiRec still works better than TiMiRec except for Recall@10 in Beauty.

D. Hyper-Parameter Discussion (RQ3)

We evaluated the following three key hyper-parameters: the number of user interests K, the trade-off parameter λ between the centrality and dispersion of user interaction, and the number of the prompt embeddings N_p.

1) The Number of User Interests K: A larger number of user interests K indicates that more interests the user has. In this experiment, we fixed the other hyper-parameters to their default values and tested K from [1,2,3,4,5]. Without losing generality, we reported the Recall@10 and NDCG@10 in Fig. 4. The results show that, as the number of user interests K increases, the model performance first increases until it achieves the best performance with the most suitable K, and then decreases. This suggests that learning multiple interests for a single user is beneficial, but having too many interests can degrade recommendation performance. The reasons are as follows. On the one hand, a user in real-world typically has multiple interests, but not an excessive number within a single domain. On the other hand, we only selected the user's most recent 20 interactions to learn the user multi-interests, and therefore there are unlikely to be too many distinct interests represented within those 20 interactions.

2) The Trade-Off Parameter λ: The trade-off parameter λ balances the importance of the centrality and dispersion of user interactions during learning the learning of user multi-interests. In this experiment, we fixed the other hyper-parameters to their default values and tested λ from [1,2,3,4,5]. The results are shown in Fig. 4. As can be seen, the model performance first increases until achieving the best performance and then decreases. This demonstrates that involving the dispersion of the user interactions appropriately does benefit the user multi-interest learning. The most suitable λ for ML-1M and Beauty datasets are 4, while that for Movies & TV dataset is 3. This may be because that the user interactions in the ML-1M and Beauty datasets are more dispersed than those in Movies & TV, indicting that the dispersion of the user interactions plays a more important role in ML-1M and Beauty.

3) The Number of Prompt Embeddings N_p: The hyper-parameter N_p controls the number of the prompt embeddings added to the inputs of user interactions. A larger N_p increases the adaptive ability of the inputs to different learning objectives in the multi-interest extractor and aggregator. In this experiment, we fixed the other hyper-parameters to their default values and tested N_p from [1,2,3,4,5]. The Recall@10 and NDCG@10 results are shown in Fig. 4. From Fig. 4, we can see that the performance first raises along with the number of the prompt embeddings N_p increasing in all datasets. This demonstrates that it is helpful to insert appropriate prompts to make the inputs adaptive to the different learning objectives. However, the model performance decreases when N_p becomes overly large. This

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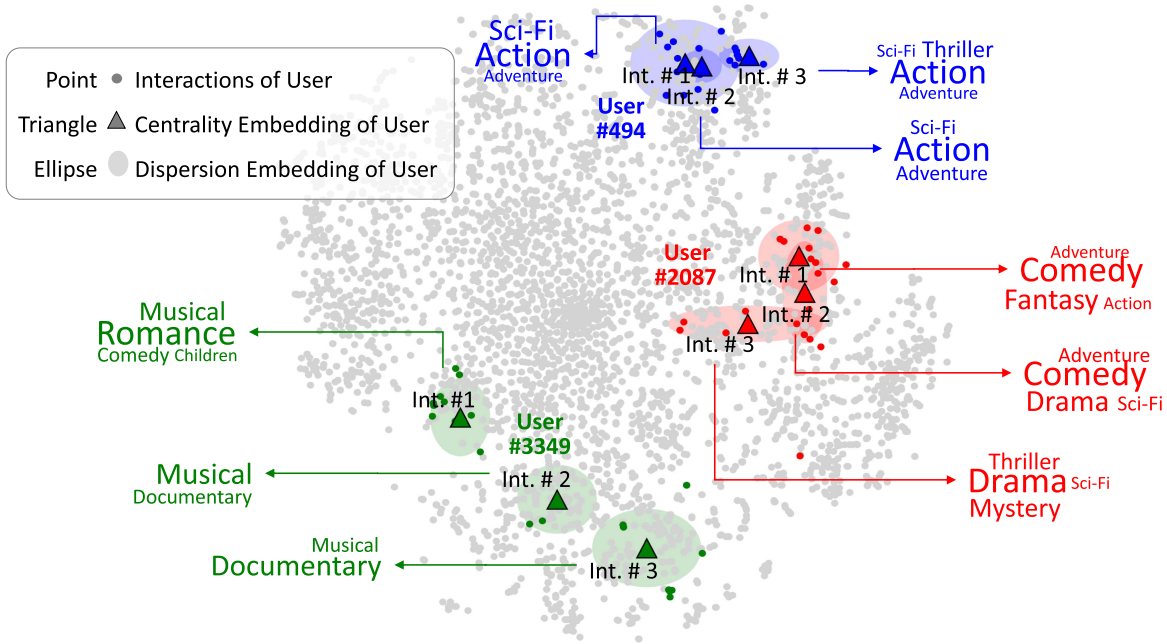

Fig. 5. Visualization of the learned user multi-interests in ML-1 M dataset with the tool of t-SNE. The grey points represent the embeddings of all items in the dataset. The interactions of three users are highlighted with different colors. The triangles represent the mean embedding of the user interests, while the ellipses refer to the variance embedding. The word cloud of user each interest generated with the item genres is annotated beside the corresponding user interest.

may be because that the multi-interest extractor and aggregator have certain connections despite having different learning objectives for the inputs. For example, the learned user multi-interest embeddings of extractor could help learn their weights in the aggregator. Therefore, excessive prompt embeddings can significantly increase the difference between the extractor and aggregator, while lessening their useful connections.

E. Visualization of the User Multi-Interests (RQ4)

To intuitively show the necessity of the centrality and dispersion of the user interactions to capture the user multi-interests, in this subsection, we visualized three examples of the learned user multi-interest embeddings. In particular, we visualized the learned item embeddings of the ML-1 M dataset with the tool of t-SNE [40] in Fig. 5. Specifically, we highlighted the historical interactions of three random users, i.e., User #494, User #2087, and User #3349 in blue, red, and green, respectively. To visualize the learned user multi-interest embeddings, we utilized the t-SNE tool to map the learned each mean vector and each variance vector into a two-dimensional vector, respectively. Each mapped mean vector in the two-dimensional space is drawn with a triangle, while the corresponding mapped variance vector is drawn with a ellipse around the triangle. Additionally, to obtain a deeper insight of the learned user multi-interests, we annotated the genres of the items belonging to each user interest by the word clouds to help understand the semantics of the three users' multi-interests. The larger the font size, the more frequently the genre occurs in the interest. From Fig. 5, we have the following observations:

- The user interests are indeed diverse and multi-faceted, making it helpful to learn multiple interest embeddings for each user to capture their multi-interests. As can be seen, the interactions of green User #3349 can be clearly divided into three groups. Checking the word clouds of item genres in the groups, we found that each group of items has clearly common genres, while different groups have different genres. Specifically, the most frequent genres of the Interest #1, #2, and #3 of the green User #3349 are Romance, Musical, and Documentary, respectively.
- Although we have learned a fixed number of interests for each user, the proposed PoMRec can deal with cases where a user has fewer interests than the fixed number. For example, as can be seen in Fig. 5, the interactions of the blue User #494 are mostly gathered in one area. In such case, the learned mean embeddings, i.e., blue triangles, of the user multi-interests are close to each other, and the variance embeddings, i.e., blue ellipses, are almost overlapping. In addition, checking the word clouds of genres in interests of the blue User #494, we found that these three interests consist of mostly movies with genres of Action. Accordingly, we can infer that this user essentially has only one prominent interest in movies, i.e., the action movies. In this case, it is reasonable that the learned user multi-interests of PoMRec are close to each other.
- The dispersion of the user interactions can complement the centrality, which helps better describe the user multi-interests. For example, as for the red User #2087 in Fig. 5, the interactions are divided into three groups. Interestingly, the interactions belonging to Interest #3 are dispersed in the horizontal axis but concentrated along the vertical

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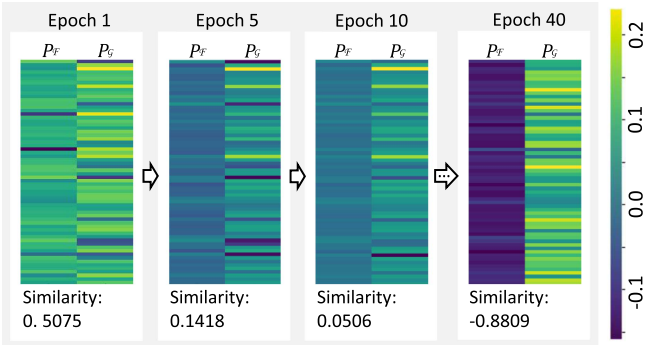

Fig. 6. Visualization of the learned prompt embeddings for multi-interest extractor and aggregator at different epochs.

axis. In such case, the mean embedding of the user interactions, i.e., the third red triangle, cannot represent the two interactions in the left. The learned variance embedding, i.e., the third red ellipse, captures the dispersion of the user interactions in Interest #3 and covers the left interactions of the user.

F. Visualization of Prompt Embedding (RQ5)

To gain a deeper understanding of the prompt embeddings, we visualized the learned prompt embeddings for the multi-interest extractor and aggregator. In particular, to intuitively evaluate the difference between the prompt embeddings for the two components, we set the number of prompt embeddings N_p to 1 and displayed the learned embeddings in different epochs with heat maps in Fig. 6, where the deeper blue color refers to a smaller value and the light yellow refers to a higher value. $\boldsymbol{P}_{\mathcal{F}}$ and $\boldsymbol{P}_{\mathcal{G}}$ are the prompt embeddings for the extractor and aggregator, respectively. We additionally calculated the inner product between $\boldsymbol{P}_{\mathcal{F}}$ and $\boldsymbol{P}_{\mathcal{G}}$ to intuitively explore the similarity. As can be observed, the heat maps and the similarity scores indicate that the prompt embeddings of the two components are initially similar during the early epochs. This is because that both prompt embeddings are randomly initialized using the same initializer, whereby they are similar at the beginning. As training progresses, the two prompt embeddings gradually diverge, with their similarity dropping below zero. This demonstrates that the prompt embeddings can be learned in a representative manner and effectively capture the different learning objectives of the extractor and aggregator.

V. Conclusion and Future Work

In this paper, we propose a prompt-based multi-interest learning method (PoMRec) for the sequential recommendation. In particular, we insert specific prompts into the user interactions to make them adaptive to the different learning objectives of the multi-interest extractor and aggregator. Moreover, we learn user multi-interest embeddings with not only the centrality of the user interactions but also their dispersion, which could comprehensively capture the user interests. Extensive experiments on three public datasets have demonstrated the effectiveness of the proposed PoMRec. In particular, we have found that although the multi-interest extractor and aggregator have their own learning objectives, they still share certain connections. Therefore, excessive specific prompts can introduce much differences between the two modules, while reducing their connections and hence negatively impacting the model performance. Nevertheless, the current PoMRec only utilizes the user interaction data, while overlooking the valuable information in item multimodal features. In the future, we plan to devise a multimodal multi-interest learning method to enhance the recommendation.

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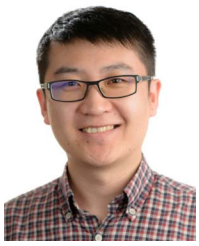

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